SoK: Evaluating LLM and Agent Side-Channel Defenses From Evidence Audits to Real-Agent Measurements

***Abstract*—Encryption conceals content while exposing the structure of agent workflows. We systematize 82 study families through 39 source-bound result records, separating the observer, execution boundary and evaluation conditions that determine whether defense evidence transfers. The synthesis identifies three consequential errors: attributing upstream protection to a plaintext-aware downstream emitter, treating discarded read cover as safe state-changing cover, and substituting one mechanism's utility or cost for another's. A populated family map and applicability-aware audit make these errors inspectable at their original source locations. Measurements on repeated agent trajectories from 53 customers test the resulting comparison obligations. Action-category localization, timing attacks and matched traffic-shaping controls expose both residual inference and costly protection boundaries; reductions in a fitted attack suite do not establish a security guarantee. The resulting contribution is a reusable evidence base and a set of demonstrated transfer failures that guide which published results can support a deployment decision, which comparisons require new measurements, and where inexpensive transport shaping matches more costly mechanisms on selected measured point metrics.**

1. Introduction

Encryption conceals message content, yet LLM engines and agents expose packet sizes, cache timing, visited domains and tool sequences [1], [2], [3], [4], [5], [6]. A defense result applies to a particular receiver, execution boundary and workload. Reusing a headline score across a frontend, dispatcher or tool provider can change both its protected signal and its cost object.

Our SoK uses the *evaluated result record* to make these transfers inspectable. Separating Dangol's stored-capture and live-protocol results prevents subtracting unmatched scores; separating Early Bird's KV-prefix and semantic-cache mechanisms prevents pricing one with the other's overhead. Combining signal origin with observer access and threat polarity explains why read-only privacy cover and state-changing bot evasion have different utility consequences. These are systematization findings: they emerge from binding evidence across records, rather than from proposing a new defense component.

Contributions.

- **A populated evidence framework.** A complete map of 58 core and 24 adjacent families, 39 source-bound records and a nine-item audit of 20 mitigation families. The framework identifies which comparisons are compatible, incompatible or unresolved.
- **Three concrete failures of evidence transfer.** Frontend re-emission defeats an upstream-only protection claim; discarded retrieval cover does not justify live action cover; stored/live measurements and component costs cannot be interchanged. The record links show exactly which published comparisons fail and which positive examples bind the required evidence.
- **Measured boundary checks on a common agent–tool chain.** Controlled streaming/state tests and repeated agent trajectories from 53 customers expose adaptation rebounds, state side effects and timing leakage. Post hoc customer-grouped analysis supplies descriptive comparisons, including cases where inexpensive transport adaptations match costly shadow shaping. Our own utility evidence is assessed under the same boundary requirements as the literature.

Shadow isolation, decoys, constant-rate tunnels and bucket padding have prior art. They are experimental probes of transfer claims here; we do not propose shadow shaping as a superior deployable defense. The comparison principles are established. The new synthesis is their application to source-bound records, including misleading within-paper comparisons that a paper-level summary would conceal. Table 1 gives the four research answers.

Scope of claims. This is a record-level SoK with descriptive mechanism checks. Cancellation/return are reference-action categories, not latent customer attributes. Repeated runs characterize one known retail cohort; 53 customers, rather than the number of runs, limit population inference. Neither attack reduction nor structural state preservation certifies security or unchanged planner utility.

Figure 1 puts trace and action paths on the same agent–tool chain. Classical data processing motivates the receiving-view check; it is not a new theorem. A detailed decision flow is retained with the supplementary methods.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions: which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems (RQ1); how

TABLE 1. RESEARCH QUESTIONS, SYSTEMATIZATION FINDINGS AND EVIDENCE. EXPERIMENTAL FOLLOW-UPS TEST THE COMPARISON RULES; THE SOK CONTRIBUTION IS THE RECORD-LEVEL SYNTHESIS.

Question	Finding	Evidence in the main paper
RQ1: targets and observables	A common signal can serve confidentiality, attribution, evasion or authorized diagnostics; polarity changes the protected interest.	Complete core-family map (§4, Table 6).
RQ2: origin versus access	B-size cover does not bind A-call order; plaintext re-emission creates a fresh B trace; shared A-call projections do not establish unchanged state. These record-level distinctions correct cross-paper transfers, not a new theorem.	NetEcho/SplitDecoy, WRA/AHB and dispatcher read/write records (§5); E5–E7.
RQ3: portability	A task holdout does not establish topology, hardware or model transfer; mixed configurations can preserve shortcuts.	Generator-specific holdouts (§7); customer-grouped analysis (§8.5).
RQ4: protection, state and cost	Bind inference, utility and cost to one scoped operating point. The 20-family audit gives R8 Bound 3/17 for component-or-task utility and 2/16 under a task-only sensitivity.	Twenty-family R1–R9 matrix; E9–E14 leakage/state/cost contrasts (§8). Full utility under timing-dependent replanning remains unestablished.

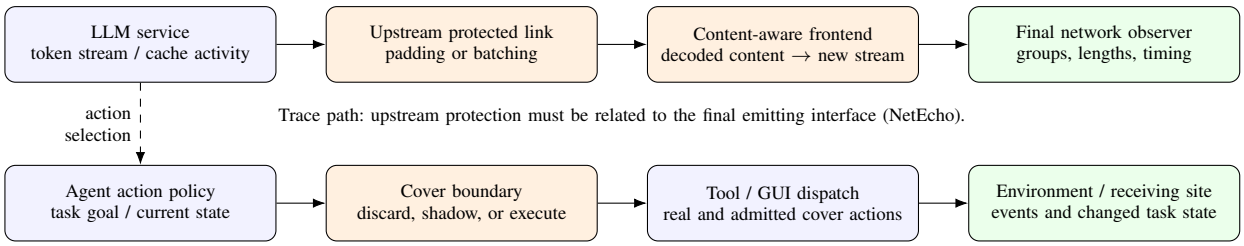


Figure 1. Observation and execution boundaries. E5 tests streaming, E6 state scope, E7 encrypted composition, E9 frozen-calibration real-agent trajectories, and E13 live tool timing with complete tunnel controls.

signal generation origin relates to receiver observation permissions (RQ2); which holdouts establish portability across tasks, configurations and hardware (RQ3); and which evidence binds workflow protection, logical state preservation and resource cost, distinguishing these from full end-to-end planner utility (RQ4).

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [7], [8], [9], [10], [11]. Table 2 states the observable eligibility rules.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap.

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

WRA protects research intent (I); UI Traces attributes agents (II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [12], [13]. These roles are mapped per core family, alongside generating and receiving layers, in Table 6.

2.2. Corpus construction and reconciliation

We construct a purposive corpus through 40 seeds, venue-targeted proceedings screening and one-hop citation snowballing, supplemented by database and bulk keyword searches. The supplementary search returned 615 occurrences and 565 unique records; auditing 204 exclusions rerouted four records, yielding 15 new core and seven adjacent families. The final collection has 58 core and 24 adjacent families. The Boolean queries overlap 25/60 previously indexed families (41.7%); this is a warning about query coverage of a known seed set, not an estimate of field-wide recall. The other discovery routes contribute known families missed by those queries. Findings describe this assembled corpus, not prevalence across all literature (search accounting: S1/S10).

2.3. Coding procedure and consistency

AI assistance produced initial schema-guided field drafts. Authors checked and corrected all 39 result records and the 180 archived audit cells against source section/table/figure/paragraph locators, and reviewed all 82 family classifications using titles, abstracts and experimental sections. These version-bound locators, rather than an agreement coefficient, support inspection of the claims. The archived draft-to-author-verification agreement is non-independent; its historical statistics and all disagreements are retained as process diagnostics in S8/S18. No independent human inter-rater reliability is claimed. Later AI-assisted source reconciliation is separately identified and never promoted to an independent human rating.

SoK Takeaway 1 (Map evidence to the receiving deployment). A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

3. Result Records, Comparability Rules and Validity

The atomic unit of our repository is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0 vs. D_2), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A study family contributes multiple records whenever it evaluates distinct execution modes (e.g., stored capture vs. live loopback in Dangol [14]), distinct mechanisms (e.g., K-prefix KV simulation vs. GPTCache semantic de-identification in Early Bird [3]), or separately scoped performance benchmarks (e.g., SafeKV [15] and OTRO [16]). Across 77 cited entries, the bibliography labels 49 arXiv-only, 21 venue-labelled (5 also cite a preprint), 1 book, 2 standards and 4 vendor documents; these are citation forms, not peer-review certifications (Supplement S18).

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

Indexing schema and comparison contracts. The mnemonic SCOPE (Secret/target, Channel, Observer, Pipeline, Exploitation goal) names the target-and-channel columns of our index, paired with evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (NR-R), unavailable (NR-A) or unverified (NR-V). Table 5 illustrates a complete record-to-judgment lineage.

Rule-application consistency check. Twenty-four author-constructed cases exercise Compatible/Incompatible/Unresolved rules. Their construction-time agreement checks implementation consistency; they do not estimate decision accuracy on external cases. Supplement

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [14].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility NR-R; precise paired live-refit comparison NR-V.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

S7 preserves every case and outcome. The seven mechanism checks recorded before E5–E7 have the same engineering role, rather than serving as independent validation of the framework.

4. Corpus Map Across 82 Study Families

4.1. Origins, receiving projections and threat polarities

Across all 58 core families in our unified repository, 38 study inherited runtime, serving or hardware channels (Origin B) and 20 study agent-workflow channels arising from orchestration, tool dispatch, browser navigation and multi-agent protocols (Origin A; Table 6). Signal generation origin is distinct from both receiver location and threat polarity. The full map makes these distinctions visible for each core family.

Cross-origin synthesis. Origin B includes token/packet lengths, cache timing and co-resident hardware channels; Origin A includes visited domains, tool dispatch and browser actions. Receiver permissions cut across these origins: a remote network observer, a same-site script and a privileged diagnostic hook expose different views even when they share an attribution target. Thus Origin A/B cannot substitute for Types I–IV or observer permissions. WRA's read-only privacy decoys, AHB's stateful evasion and SplitDecoy's exposed return path illustrate transfers that fail when these axes are collapsed [6], [12], [29]. The map retains every core family rather than using these examples as a substitute for coverage.

4.2. Relationship to prior systematizations

Prior surveys establish compound-system threat models and confidential-computing boundaries. Banerjee et al. categorize cross-layer compound AI threats under MITRE tactics; Forough et al. distinguish TEE isolation boundaries from side-channel resistance and note that inference protections need not cover agent memory or action coordination [64], [65]. Kotevska et al. propose a six-element privacy contract (asset, observer, channel, permitted disclosure, guarantee, evidence) with composition conditions framed as formal conjectures [66], while classical website-fingerprinting works establish bandwidth/latency bounds for transport shaping (Tamaraw, WTF-PAD, FRONT, Walkie-Talkie) [67], [68], [69], [70]. Our distinct contribution is the populated 82-family / 39-record repository, the nine-item comparability audit, and mechanism checks recorded before E5–E7 (S12) for the transfers it flags, using loopback TLS replays and a pinned stateful tool simulator. Website-fingerprinting defenses already show that per-packet padding leaves counts and that constant-shape shaping is expensive [67], [68]. E5 measures how that known property invalidates a specific cost transfer for LLM token streams, without claiming it as new.

5. Three Literature Case Studies

Record-level synthesis resolves three deployment decisions: whether upstream shaping survives another workflow hop, whether cover preserves the relevant state, and whether adaptive evaluation and cost describe the same mechanism. The following cases provide the source-local evidence for each judgment.

5.1. Case 1: bind the final target-bearing emitter

The downstream re-emission boundary. NetEcho's upstream packet-size shaping is an Origin B result; it does not bind the Origin A tool-call sequence. Its plaintext-aware frontend then emits a new Origin B trace (grouped token counts or character lengths), reopening a size channel after upstream protection [60]. SplitDecoy is a separate return-path record: the cloud sees nonzero gradient row support on every tested frame across nine seeds [29, Sections 5.1–5.3]. The inference is a changed receiver boundary, not one transferable score across B and A.

The standard data processing inequality applies when $S \to Z \to Y$ is a Markov chain: target-oblivious post-processing cannot increase $I(S;Z)$ [71, Theorem 2.8.1]. A plaintext-aware frontend or unprotected backward computation changes that premise. The systematization insight is identifying which published records cross this boundary.

5.2. Case 2: resolve the WRA versus AHB cover asymmetry

Two axes of asymmetry: threat polarity and action risk. A comparison that groups WRA's decoy requests [6] and AHB's humanized actions [12] together as "cover traffic" can create an apparent paradox: WRA reports a clean privacy improvement (one concurrent decoy reduces

TABLE 6. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; MULTIPLE LAYER/TYPE LABELS DESCRIBE DIFFERENT COMPONENTS OR OBJECTIVES WITHIN THE FAMILY.

Core family	$G \to O$	Type	Core family	$G \to O$	Type
ArchSnoop [17]	$1 \to 1$	II	PromptPeek [2]	$2 \to 4$	I
Behind Bars [18]	$1 \to 1$	I	Remote keylogging [1]	$2 \to 3$	I
Edge EM leakage [19]	$1 \to 1$	I	Remote timing [20]	$2 \to 3$	I
Energon [21]	$1 \to 1$	II	Routing metadata [22]	$2 \to 4$	I
I Know What You Said [23]	$1 \to 1$	I	SafeKV [15]	$2 \to 4$	I
MoEcho [24]	$1 \to 1$	I	Speculative decoding [25]	$2 \to 3$	I
PAGE404 [26]	$1 \to 1$	I	SpliceLeak [27]	$2 \to 4$	I
Spill the Beans [28]	$1 \to 1$	I	SplitDecoy [29]	$2 \to 2$	I
Detokenization Leaks [30]	$1, 2 \to 1$	I	Time Will Tell [31]	$2 \to 4$	I
LLM-serving RCA [32]	$1, 2 \to 2$	IV	Whisper Leak [33]	$2 \to 3$	I
LLM-WFIN [34]	$1, 2 \to 1, 3$	I	MIRAGE [35]	$2, 5 \to 2$	IV
OTRO [16]	$1, 2 \to 1$	I	Cloud traffic benchmark [36]	$2, 5, 8 \to 3$	I, II
Shape and Substance [37]	$1, 2 \to 1$	I	Web-agent unmasking [38]	$3, 7 \to 3, 7$	II, III
SparLeak [39]	$1, 2 \to 1$	I	LumiMAS [40]	$4, 5 \to 4, 6$	IV
SparSEEty [41]	$1, 2 \to 1$	I	AgentPrint [4]	$5 \to 3$	I, II
TDXRay [42]	$1, 2 \to 1$	I	Cycle detection [43]	$5 \to 4, 6$	IV
ClawGuard [13]	$1, 5 \to 1$	IV	AgentProv [44]	$5, 6 \to 6$	II
Cache contention [45]	$2 \to 4$	I	Lidar [46]	$5, 6 \to 6$	II
CachePrune [47]	$2 \to 4$	I	ProcGrep [48]	$5, 6 \to 4, 6$	II
Early Bird [3]	$2 \to 4$	I	SIGLEAK [49]	$5, 6 \to 4, 6$	I
InputSnatch [50]	$2 \to 4$	I	Campaign attribution [51]	$5, 6, 8 \to 6, 8$	II
KVGov [52]	$2 \to 2, 4$	I	Dangol workflow [14]	$5, 6, 8 \to 3, 6, 8$	I
Leaky Language Models [53]	$2 \to 4$	I	AgentTell [54]	$5, 7 \to 7$	I
LLMs Have Rhythm [55]	$2 \to 3$	II	AHB [12]	$5, 7 \to 7$	III
ML privacy channels [56]	$2 \to 4$	I	FP-Agent [57]	$5, 7 \to 7$	II
MoE prompt stealing [58]	$2 \to 4$	I	PersonaFingerprint [59]	$5, 7 \to 3$	I
NetEcho [60]	$2 \to 3$	I	UI Traces [5]	$5, 7 \to 7$	II
Prompt Echo [61]	$2 \to 3$	I	Whose Agent? [62]	$5, 7 \to 7$	II
Prompt-cache audit [63]	$2 \to 4$	I	WRA [6]	$5, 7 \to 3$	I

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and the normalization of prose-only layer descriptions are in the artifact.

functional-intent reconstruction E_{func} from 0.770 to 0.610), whereas AHB reports severe task collapse (Trip Planning success falls from 0.75 Raw to 0.15 under combined `Online+Fake` humanization).

Both records originate in Origin A actions, but polarity and state effects differ: WRA's concurrent decoy retrievals are discarded reads for Type I confidentiality, while AHB's live GUI events evade a Type III platform detector and can navigate or submit forms. Their utility outcomes are not contradictory, and AHB's missing condition-specific denominators preclude a matched causal effect. Origin A alone predicts neither receiver permissions nor whether an action mutates protected state.

Terminal-state reward does not establish dispatch-time authorization: an unconfirmed write could happen to produce the task's intended final state. For stateful tools, audit the real execution path and protected account/database, not only the terminal hash. E6 tests this distinction with actual simulator mutations.

5.3. Case 3: bind adaptive retraining and cost objects

The $D_0 \to D_2$ rebound and component-cost boundaries. A matched adaptation contrast compares an undefended-data-trained attacker D_0 and a defended-data-trained, defense-aware attacker D_2 on identical defended tests. UI Traces' timestamp-delay refitting and Dangol's stored/live distinction show why receiver and run provenance must accompany the score [5], [14]. E5/E6 supply direct contrasts where D_0 reaches chance and D_2 recovers the signal; E14 expands the attacker families.

On the cost side, Early Bird's 6ms/4% GPTCache overhead does not price K-prefix isolation; KVGov's $1.3\,\mu$s HMAC microcost omits lost cache reuse; and SpliceDefense's QCP cost does not price combined CTBF+QCP [3], [52], [27]. CachePrune binds attack suppression, QA/summarization utility and throughput; OTRO reports access-count leakage conditional on byte length and separately scoped TTFT/padding cost [47], [16]. Even within Origin A, identical call order is not a state guarantee: read/write traces can retain the same dispatch features while writes alter the account or database. The state claim therefore requires execution-path and final-state checks (E6), not

the Origin label or trace score alone.

6. Comparability Audit Across All 58 Core Families

6.1. Unified 58-core disposition and 39-record ledger

The corpus partitions as $82 = 58$ core $+24$ adjacent families. Within the core, $58 = 20$ mitigation $+6$ contextual-record $+32$ map-only families; the result ledger contains $39 = 33 + 6$ records, respectively, from the first two groups. The six are SIGLEAK, AgentTell, FP-Agent, Behind Bars, LLMs Have Rhythm and ML privacy channels [49], [54], [57], [18], [55], [56]. They supply fixed-guardrail, access-control, interference, VPN or filter-origin contrasts rather than the audited mitigation claim. The remaining 32 comprise 22 without a scoped mitigation evaluation, five diagnostics, four unbound assertions and SplitDecoy's boundary case. S18 maps every family and record. Table 7 groups mechanisms; Table 8 reports the 20-family audit.

6.2. Within-family and within-cohort record comparisons

While the machine-readable audit retains all $\binom{33}{2} = 528$ baseline pairs for completeness, 426 pairs cross unrelated mechanism cohorts and are outside scope by construction. The 13 within-family pairs are included in the 102 within-cohort pairs: $528 = 426 + 102$, with no additional subtraction of 13. Focusing on the **13 within-family pairs** (multiple records inside the same paper) and **102 within-cohort related pairs** (66 traffic-envelope, 28 cache-service, 3 behavioral-humanization, 3 generator-fingerprint, 1 local-memory, 1 fixed-instruction; Supplement S8) reveals why paper-level comparisons fail:

- Even within the **13 same-family pairs** (plus 2 same-family pairs in the additions: `CACHEPRUNE-selective` vs. `CACHEPRUNE-context` and `OTRO-unpadded` vs. `OTRO-epochpad`), nearly half (6/13 baseline; 7/15 full) measure different primitives, targets or cost objects (e.g., Early Bird's K-prefix oracle simulation vs. its GPTCache semantic prototype; SpliceDefense's CTBF signal suppression vs. QCP-only prefill cost; OTRO's unpadded TTFT vs. derived epoch-padding cost).
- Among the seven aligned-outcome same-family pairs, four evaluate different collection or execution populations (e.g., Dangol's stored-capture feature ablation at 0.17 accuracy vs. live loopback OHTTP at 0.30 accuracy) and three lack a verified matched-test binding, so none licenses cross-record subtraction as a single paired deployment effect.
- Across the **102 related-cohort pairs**, only 16 (15.7%) support a source-qualified descriptive comparison, whereas 80 (78.4%) exhibit positively established receiver, target or cost-object mismatches and 6 (5.9%) remain unresolved (the supplied 50-pair subsample, not all 102 pairs, has 94.0% agreement across its 150 judgments, $\kappa = 0.916$).

6.3. Nine-item split reporting audit across 20 mitigation families

Table 8 reports all 180 family/item bindings. R1–R9 are evidence conditions for reusing a result, not paper-quality or safety scores; applicability follows the scoped claim, and Bound means at least one family record meets it. R4 asks whether a fitted/calibrated attacker is evaluated on protected data; R5 asks whether clean/adapted fits share a defended test. Their 11 applicable families include NR-R (1 each) and NR-V (5 for R4; 6 for R5): unreported or unverified bindings are not failed experiments. R7–R9 record inference, utility and cost.

For Dangol, R5 is Bound for in-fold stored-capture experiments; the live-refit paired comparison in Table 5 remains NR-V. These are different records, so we retain the family count of two and explicitly restrict its meaning.

SoK Takeaway 2 (Evidence needed for reuse). The audit distinguishes component-or-task utility (Bound 3/17 applicable families) from deployment-task utility (2/16 when TDXRay's lookup microbenchmark is treated as inapplicable); these are scope sensitivities, not quality ranks. R5 binds matched adaptation for 2/11 applicable families; a narrower learned-adaptive-attack denominator excluding Shape and Substance is 2/10. CachePrune binds inference, utility and cost together; it does not thereby establish adaptation. Full cell dispositions and reasons are in Supplement S18.

7. Match Transfer Claims to the Generator

7.1. Different tests rule out different explanations

A fingerprint is generated by a model interacting with tasks, prompts, tools, browser state and serving infrastructure. A holdout establishes portability along the axis it changes: new tasks, browser configurations, network topologies and hardware are distinct deployment tests. Supplement S2 maps each dependency to its supporting holdout.

UI Traces fixes a browser/action harness while testing tasks and environments. FP-Agent combines behavior with static browser attributes; shared visitor proxies are not authenticated independent people or hosts [5], [57]. Task separation can remove memorization of those task instances while retaining configuration identifiers. This is not proof of an invalid original split; it bounds portability. The traffic benchmark separately tests generating topology, so its shift is a different test rather than a stronger score on the same estimand [36].

LeakyLMs illustrates the boundary for system-property rather than agent identity. Its deployment probe observes streaming events through an ordinary client API. Its architecture branch additionally assumes known GPU type, same-class calibration and single-GPU execution [53]. Remote

TABLE 7. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [60], [33].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains NR-R.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [14].	The co-located loopback setting yields 0.30 recovery at about $13\times$ bytes and $11\times$ latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are NR-R.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [12].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [15], [52], [47].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median 1.3 μs prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [27].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and < 5ms/$< 1\%$ prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [42], [23], [16].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

architecture inference therefore does not establish hardware-agnostic inference. The joint search in its remote example ranks the true configuration 28/2068, distinct from a conditional parameter search or a near match; those outcomes must not be collapsed into universal exact recovery. Source-reported local architecture generalization and remote deployment signatures remain positive evidence under their own assumptions.

7.2. Representation increment is not portability

Within a fixed observer and test population, coarse counts versus sequence features test whether the latter improve a specified predictor. WRA already uses ordered domains before adding timing; Kang changes class sets across some conditions [6], [62]. These test different contrasts from an order-only effect. The supplementary archive example fixes labels and units and finds a 0.3447-nat sequence increment under archive-receiver ordering. Its small common-task control leaves task-distribution explanations unresolved (Supplement S4).

Adjacent coding-agent PR attribution depends on repositories, submission pathways and artifact generation. Its mixed-feature result does not isolate an auxiliary-only C3 view; inspected code computes weighted F1 and reduces features before cross-validation, but the exact paper-run binding remains unverified [7]. We neither rewrite its headline score nor quantify optimism. WildAuth is adjacent under the same rule; input/partition ambiguities do not supply a model-independent response-only privacy claim [8].

7.3. Resulting claim hierarchy

Task-specific identification, task-disjoint identification in a fixed harness, configuration/topology transfer and hardware-calibrated architecture inference require different evidence. Origin A/B labels locate generating layers; they do not establish a receiver view or cross-layer equivalence. Our case synthesis uses them to correct three tempting transfers: upstream B-size cover does not bind A-call order, plaintext re-emission creates a fresh B trace, and an unchanged A-call projection does not prove state invariance. These are record-grounded boundary findings, not new theorems. For portability claims, change the relevant generator while preserving target labels and receiver contract; for representation increments, compare identical units and disclose predictor/selection restrictions.

8. Controlled Transfer Tests

The experiments connect observation, execution and matched evaluation. We retain artifact identifiers: E1/E3

TABLE 8. PER-FAMILY EVIDENCE BINDINGS FOR ALL 20 EVALUATED-MITIGATION FAMILIES. R1–R9 ARE SCOPED REUSE ITEMS, NOT OVERALL PAPER SCORES.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [5]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	U	–
WRA [6]	**B**	**P**	**B**	–	–	**P**	**B**	**P**	U
NetEcho [60]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	U
Early Bird [3]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	**B**
Whisper Leak [33]	**B**	**B**	**P**	?	?	**B**	**B**	U	**B**
SpliceLeak [27]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [31]	**B**	**B**	**B**	?	?	**B**	**B**	U	**B**
I Know What You Said [23]	**B**	**B**	**B**	–	–	**B**	U	–	**B**
Speculative decoding [25]	**B**	**B**	**P**	?	?	**B**	**B**	–	**B**
Dangol workflow [14]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	U	**B**
Cloud traffic benchmark [36]	**B**	**P**	**B**	?	?	**P**	**B**	**P**	U
Remote timing [20]	**B**	**B**	**B**	–	–	**B**	**B**	U	**P**
TDXRay [42]	**B**	**B**	**B**	–	–	**B**	U	**B**	**B**
SafeKV [15]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
AHB [12]	**B**	**B**	**B**	**P**	?	**B**	**B**	**B**	U
KVGov [52]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
CachePrune [47]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [16]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
SparLeak [39]	**B**	**B**	**B**	?	?	**B**	**B**	**P**	U
Shape and Substance [37]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	20	17	17	4	2	18	17	3	11
Applicable	20	20	20	11	11	20	20	17	19

B: bound; **P**: partial; U: unreported in inspected scope; ?: unverified binding (11 cells); –: inapplicable. R1 target/polarity; R2 receiver; R3 placement/execution; R4 defended-condition fit; R5 matched adaptation; R6 workload/generator shifts; R7 inference; R8 utility; R9 cost. A bound cell is supported by at least one scoped record. Dangol R5 concerns stored capture; its live-refit binding remains unverified. CachePrune jointly binds R7–R9 at one operating point. Source follow-up resolves three former ? cells: cloud-traffic R3 to B; Shape and Substance R4/R5 to U (S8).

cover buffered transport and adaptive fitting (§8.1; S11), E2 tests state-preserving cover (§8.2; S9/S11), E5–E7 isolate streaming and dispatch mechanisms, E8 tests stand-in planners (S13), E9–E12 analyze real-agent trajectories, E13 measures live tool timing/full tunnels, E14 strengthens attacks and compares classical WF adaptations, and E15 adds same-provider model trajectories. S18 gives the full concordance, including unused E4. We use *undefended* for transport without a program defense, and *standard prompt* for the prompt arm called `baseline` in immutable data. Artifact keys `raw`/`baseline` are retained for reproducibility. Seven author-constructed mechanism checks were recorded before E5–E7 (S12); they are engineering checks rather than a sample of external review decisions.

The original E6/E7/E9 protocol uses an author-chosen *evaluated-suite criterion* when maximum BA has an upper 95% bound below 0.60 and maximum AUROC is below 0.65. It is an engineering screening rule, not a cryptographic definition or a field-standard acceptable-risk threshold. We retain it for protocol accountability, not as the paper's contribution. A measured reduction alone cannot demonstrate that a deployment is protected: a lower BA bound above 0.60 indicates above-threshold leakage, while an interval crossing 0.60 leaves the absolute threshold inconclusive. We recompute the suite maximum within each of 2,000 common bootstrap draws: tasks for E6/E7 and customers for E9, retaining their tasks and repeats together. Intervals condition on fitted predictions and do not cover unknown attackers.

TABLE 9. DISTINCT ESTIMANDS, NOT INTERCHANGEABLE RESULTS. STANDARD-PROMPT PAIRS: CANCEL/RETURN BA (%). COUNTS COVER BOTH PROMPTS; BYTE RATIOS PRICE JOINT CONSTANT SHAPING AGAINST EACH STUDY'S OWN UNDEFENDED SERIALIZATION.

Study	Observation / fit	Raw BA	Bytes×
E9	Replay lengths/order; frozen suite	97.2/94.0	15.39
E10	Analytic lengths/order; 11-model cross-fit	91.7/85.5	19.76
E13/14	New RPC envelope; lengths/order/time	91.7/86.4	15.11

E9/E13: 276 test episodes, 24 customers, 5/7 positive customers. E10: 690 episodes, 53 customers, 14/18 positive customers. Each prompt has half the episodes. E10 fold calibration yields larger templates (5.04× calls); E9/E13 use 3.83×. The 690 runs are repeated execution coverage, not 690 independent customers.

A byte-transparent relay captures post-handshake TLS application records on loopback connections. TLS 1.3 session tickets are disabled. E5–E8 use saved text or stand-in planners; E9 invokes a real model and executes its structured actions in the local public retail simulator. No production account or tool is modified.

8.1. Transport: buffered and streaming emitters

The buffered tests use brief/expanded saved responses for 24 topics (12 training, 12 test). A two-link TLS replay remains 100% separable on the terminal link when only upstream traffic is padded; padding both reduces verbosity inference to 50%, reproducing NetEcho's mechanism [60]. For TLS 1.2, a training-fitted buffered cap reaches 50% at 2.18× bytes; Table 10 uses TLS 1.3 and its own denominator (2.19×). Defended-data fitting recovers 13.5–50.0 points under offsets, buckets and random tails (S11).

E5 emits the same 48 texts as per-token server-sent events over TLS 1.2/1.3, reflecting the streaming surface [1], [33]. Per-event padding preserves token count: verbosity remains 100% separable. Eight-token batching cuts bytes to 0.21× and clean-trained BA to 50%, but refitting recovers 100%. Constant shape reaches 50% at 2.42× bytes and 208 versus 105 token intervals; two overflowing test outputs remain identifiable. Table 10 gives all emitter, attacker and cost cells. A buffered cap delays the first byte until generation ends, so its cost cannot price streaming protection.

8.2. Execution: state scope and randomized cover

State scope. In the 115 public retail programs, same-item modification changes status and appends a refund even when item and price are unchanged (173/291 pre-insertion units). State preservation $f(s, a) = s$ therefore differs from idempotence $f(f(s, a), a) = f(s, a)$. Cancelling another user's order changes the shared database. Shadow cover separates an observable call from its live effects: authorized

TABLE 10. E5: THE SAME 48 SAVED RESPONSES EMITTED BY A BUFFERED OR A TOKEN-STREAMING FRONTEND (TLS 1.3 LOOPBACK CAPTURES). BYTES ARE RELATIVE TO THE UNPADDED EMITTER OF THE SAME KIND. TTFB/DONE: FIRST AND LAST DELIVERED BYTE IN TOKEN INTERVALS (ACCOUNTING PROXY). ACCURACIES ON 24 HELD-OUT OUTPUTS (12 TOPICS); ID: CLOSED-WORLD TOP-1 IDENTIFICATION AMONG 48 RESPONSES (CHANCE 2.1%).

				T1 acc.		T3	
Emitter / framing	Bytes×	TTFB	Done	D_0	D_2	D_2	ID
Buffered, unpadded	1.00	105	105	100.0	100.0	91.7	87.5
Buffered, training cap	2.19	105	105	50.0	50.0	50.0	2.1
Stream, per token	1.00	1	105	100.0	100.0	87.5	100.0
+ random event pad	1.29	1	105	100.0	100.0	87.9	62.3
+ constant event size	1.23	1	105	100.0	100.0	87.5	58.3
8-token batches, const.	0.21	8	105	50.0	100.0	87.5	20.8
Const. events + count cap	2.42	1	208	50.0	50.0	50.0	6.3

D_2 is the best of total-byte threshold, record-count threshold and a logistic sequence model, each fitted on defended training outputs. Random padding: 10 socket-measured draws plus an independent profiling run. TLS 1.2 gives identical accuracies and byte ratios within 0.03. Tokens: o200k_base proxy. Count cap: 208 (T1) and 88 (T3) events fitted on training outputs; 2/24 and 5/24 test outputs overflow it.

reads, noncommitting API operations or a trusted gateway's isolated executor can supply this boundary (§9.1). A known-position cover call remains removable with 100% accuracy (S11).

E6: randomized cover against policy-aware attackers. E6's dispatch observer sees ordered tool names and response lengths. Cover is sampled from other users' programs, randomly interleaved and executed on shadow state, with live execution as a control. Policies add cancellation, pad its count, interleave decoy programs or equalize every tool's count. Templates, donors and caps use all 115 reference programs: this is known-population defense calibration. Attackers use clean/defended gradient boosting and logistic regression on counts, bigrams and lengths, plus a count rule. Task-grouped 5-fold evaluation uses three seeds and cancellation/return targets (18/32 positive tasks).

Table 11 reports all dispatch policies. One cancellation drives undefended-data-trained BA to 50%, but refitting recovers 100% (full table in S12). Count equalization defeats the count rule while correlated tools yield 76.9% . Decoy programs, the read-only pattern of WRA [6], leave 83.5–91.9%. Shape equalization reaches 50.0%/54.5% at 6.7 extra calls per real call. The replay checker confirms the designed state invariant in all shadow cases; live cancellation violates it in every case. This checks implementation conformance, not live planner performance.

8.3. E7: composing execution and transport on the agent–tool link

E7 composes dispatch and transport on an encrypted agent–tool link (Figure 1). Four cover policies yield 4,140 TLS 1.2 sessions. Framing is undefended, 1024-byte buckets or constant requests/responses with overflow buckets. The evaluated network projection contains direction-tagged TLS record lengths and order; it excludes tool names and

TABLE 11. E6 MAIN DISPATCH RESULTS: ALL 115 REFERENCE PROGRAMS, THREE SEEDS, KNOWN-POPULATION DEFENSE CALIBRATION; TASK-FOLDED ATTACKERS. MAXIMUM BA (CANCEL/RETURN); EXTRA CALLS PER REAL CALL; DATABASE EQUALITY.

Policy	BA (%)	Calls+	DB eq.
No cover	100.0/100.0	0.00	100%
Cancel, $p = .5$	86.1/100.0	.10	100%
Cancel, $p = 1$, shadow	100.0/100.0	.20	100%
Cancel, $p = 1$, live	100.0/100.0	.20	0%
Cancel count $\to 2$	76.9/100.0	.35	100%
One decoy program	91.9/85.9	1.01	100%
Two decoy programs	83.5/78.1	1.98	100%
Shape equalization	50.0/54.5	6.71	100%

inter-arrival times. A profiling attacker learns size-to-tool mappings from training-fold API calls and applies dispatch attacks to decoded traces; S12 specifies caps and classifiers.

Request-only tool identification reaches 89.9/91.4% under cancellation/return folds (582 calls; majority 29.4%). Framing alone leaves bucketed return BA at 73.9%. Joint shape cover with constant framing gives 54.4/50.0% at 26.3× bytes and 7.7× calls; no condition meets the two-target criterion. S12 gives all policies, and S13 retains E8's stand-in-planner comparison.

Transport link versus dispatch boundary. Per-call framing leaves call counts and turn structure exposed. A full traffic-flow-confidentiality tunnel can instead conceal these from a passive link observer using fixed-size, constant-rate traffic [72]; protecting session duration additionally requires a workload-independent traffic horizon. Its dummy packets need not execute tools. E7/E9 test per-call framing; E13 separately measures a complete fixed-rate TLS tunnel. At a dispatch boundary that exposes calls after decapsulation, transport padding is removed and call-level cover becomes relevant. Protection then depends on hiding cover markers and downstream routing from that observer (S17).

8.4. E9: real agents and training-only defense calibration

E9 tests program transforms of autonomous model-generated trajectories under a frozen calibration boundary. Its targets are whether the task's reference program requests `cancel_pending_order` or `return_delivered_order_items`; they are fixed before execution, not inferred from the realized trajectory. They nonetheless encode intended operation types, so identifying their writes is an expected diagnostic, not evidence of latent-content leakage. GPT-6.1 Sol (low reasoning effort) plans and responds from the retail policy, public tool schemas and actual sandbox observations. A separate context of the same model simulates the user. The rules arm adds one customer-details read before order access and a read before each order write. All 115 tasks have two prompt arms and three repeats, completing the frozen 690-run protocol. All 414 training runs on 69 tasks precede freezing of a shared count template, donor pools

and framing caps; the 276 test runs cover 46 tasks from 24 disjoint customers.

Planner-visible equivalence. Let H exclude wall-clock timing. Passing real calls/responses unchanged while isolating cover state and hiding cover responses gives $\pi_{\text{plan}}(D(H)) = H$ (Appendix A). Reusing a trajectory isolates the transform from LLM sampling, but does not preserve queueing, timeouts or replanning. Prefix-only scheduling is online; trajectory collection is not repeated under each defense. The trusted executor knows cover identities and is outside the protected projection.

The dispatcher inserts training-derived cover using only past calls and flushes unmet counts at episode end. A timing-aware observer could exploit this flush or read/write service times; E9 measures the length/order projection, not a defense against that observer. Four program conditions per prompt isolate undefended transport, padding, shape cover and their combination. Training-fitted attackers evaluate dispatch and network observations across 2,760 TLS replay sessions. Every test task, failure and handoff remains in the analysis. Supplement S14 supplies the full ablations and customer-cluster intervals.

Behavioral control and frozen-split reference. Rules raise mechanical read-order adherence from 86.2% to 97.1%, while undefended intent BA remains above 93%. This read-order control is not a privacy defense. Rewards are 110/138 versus 105/138 (paired difference -3.6 points, interval $[-9.5, 1.4]$). Program conditions inherit these rewards from reused trajectories; defended planner utility is not remeasured.

The frozen joint-shaping cell has chance-level point estimates, with upper bounds 60.6%/65.5%; the absolute criterion is unconfirmed. The paired decreases are 47.2/44.0 points. Rules plus joint shaping leaves return BA at 60.6%, demonstrating sensitivity to the prompt/cohort realization. Five cancellation-positive and seven return-positive test customers limit precision. The following all-customer comparison is exploratory. E9 retains its original calibration chronology; neither analysis is a new-customer confirmation, since the full task population was available during development.

8.5. E10–E12: cross-fitting, operating points and mechanism

E10: exploratory within-cohort comparison. A post hoc customer-grouped stratified 5-fold analysis reuses all 690 runs: 53 customers, including 14 cancellation-positive and 18 return-positive customers. Each fold fits defense templates, donors, caps and attackers on the other customers only. Its analytic TLS 1.2 model matches all 2,760 measured sessions and 120 fresh captures. We expand all nine prior attacks with a trained 1D-CNN and k-FP-style forest-leaf attack; Table 13 uses the maximum over all 11. Undefended BA is 91.7%/85.5%. Recalibrating the joint shadow-count/constant-framing design reduces BA by 31.2/30.0 points (paired 95% intervals [15.3,41.0]/[14.8,39.2]); joint buckets reduce it by 34.6/32.8 points. Stronger learners preserve these reductions while increasing some defended estimates. Neither joint design establishes the original absolute criterion. The paired effects answer whether these transforms reduce this fitted suite's inference, not whether they withstand an arbitrary attacker (S21).

A fixed-prediction sensitivity check weights each customer equally within each target class. Standard-prompt joint-constant BA changes from 60.5/55.5% to 59.0/53.2%; rules-arm return changes from 59.4% to 60.4%. Removing each positive customer in turn gives standard-prompt cancellation BA from 57.3 to 62.0%. These 192 deletion checks expose cohort influence; they are not new samples or confidence intervals and omit attacker-refitting uncertainty (S23).

E11's retrospective six-setting selection remains in S15; its older-suite cost is not the current comparison.

E12: operation-category localization, not content inference. The expanded 11-attack oracle ablation (Table 14) retains 91.7%/84.9% BA in state-changing exchanges, which comprise only 19.6% of real tool calls. Removing them reduces BA to 62.8%/59.9%: paired decreases of 28.9 [10.7,41.9] and 25.6 [11.0,35.4] points. This localization is label-coupled: the target names the intended write operation. It establishes that ciphertext metadata can preserve an operation signature after names are hidden, but not that writes generally reveal unrelated private attributes. Reads retain predictive metadata. Rules reduce distinct read sequences from 74 to 62, while write sequences remain 37 versus 36. The oracle selects exchanges using true tool types, then exposes only their network lengths/order; it is a localization diagnostic, not a deployable filter.

8.6. E13: live timing, latency and a complete tunnel

E13 executes all 690 saved action trajectories through a live local tool server under six conditions: 4,140 new TLS sessions. Each real response gates the next action; real and shadow calls use separate states. The original customer-disjoint 414/276 training/test split is retained. Attackers are fitted and frozen on training captures before test collection. Ciphertext lengths, directions and arrival times measure the tool RPC chain, excluding LLM generation and WAN transit (S19).

The tunnel sends 4,096-byte cells in both directions at 2 or 5 ms target intervals, including dummy cells that execute no tools. Its training-sized 133-slot horizon remains fixed after the last response. Serial and two-in-flight shadow conditions use E9's prefix-only count policy and constant per-call framing. The attack suite includes size/order, timing, joint features and learned flush removal; all transformations use training customers only.

The clock and the distinguishable cover. For standard-prompt serial shadow cover, size/order BA is 50.0/50.7%, versus 61.4/53.4% with the original full suite. Cancellation increases by 11.4 points (paired 95% interval [3.7,19.6]). Flush-phase detection reaches 90.4/92.3% BA for serial/parallel schedules across 15,278 records from 276 test episodes (24 customers, both prompts). The distinct cover-versus-real request classifier reaches 82.5/96.1% on

TABLE 12. E9: FROZEN-CALIBRATION OPERATION-CATEGORY DIAGNOSTICS ON REAL-AGENT TRAJECTORIES. TESTING COVERS 46 TASKS FROM 24 HELD-OUT CUSTOMERS, TWO PROMPTS AND THREE REPEATS (276 EPISODES). NETWORK BA IS THE MAXIMUM OVER THE FIXED ATTACK SUITE, WITH CUSTOMER-CLUSTER 95% INTERVALS.

Condition	Cancel BA [95% CI]	Return BA [95% CI]	Reward (%)	Bytes×	Calls×
Undefended	97.2 [90.0, 100.0]	94.0 [85.8, 100.0]	79.7	1.00	1.00
Rules prompt	97.2 [90.0, 100.0]	93.5 [85.0, 99.5]	76.1	1.00	1.00
Padding only	66.1 [50.0, 83.9]	55.5 [50.0, 73.0]	79.7	4.02	1.00
Shape only	60.8 [55.4, 71.3]	51.9 [50.5, 57.8]	79.7	4.16	3.83
Joint program	50.0 [50.0, 60.6]	50.0 [50.0, 65.5]	79.7	15.39	3.83
Rules + joint	50.0 [50.0, 65.1]	60.6 [50.0, 73.3]	76.1	15.72	3.88

Paired joint-minus-undefended BA differences (percentage points): standard-prompt cancellation -47.2 [-50.0, -33.3], return -44.0 [-48.9, -27.8]. These post hoc contrasts use the same customer draws and reselect the attack-suite maximum in each condition. Joint = online shadow cover plus constant framing; padding/shape ablations use standard-prompt trajectories. Ratios use undefended transport within each prompt. Shadow isolation preserves the logical planner transcript; the four program conditions replay actual LLM trajectories over local TLS 1.2. Full methods, intervals and prompt contrasts: Supplement S14.

TABLE 13. EXPLORATORY E10 CROSS-FIT COMPARISON UNDER THE EXPANDED 11-ATTACKER UNION: ALL 690 EPISODES AND 53 CUSTOMERS. MAXIMUM NETWORK BA WITH CUSTOMER-BOOTSTRAP 95% INTERVALS. STANDARD PROMPT UNLESS MARKED RULES; BYTE/CALL RATIOS RELATIVE TO UNDEFENDED TRANSPORT. JOINT-DESIGN ABSOLUTE-CRITERION OUTCOMES REMAIN INCONCLUSIVE.

Configuration	Cancel BA (%)	Return BA (%)	B×/C×
Undefended	91.7 [82.1, 100.0]	85.5 [76.7, 93.4]	1.0/1.0
Joint constant	60.5 [52.9, 72.4]	55.5 [51.8, 64.2]	19.8/5.0
Joint buckets	57.1 [50.0, 71.5]	52.7 [50.0, 61.7]	12.4/5.0
Rules + buckets	56.9 [50.2, 71.2]	52.7 [50.0, 65.0]	12.4/5.0

Joint constant is E9's shadow-count plus constant-framing design, refit within each fold. Joint buckets changes framing to 1024-byte buckets. Every bootstrap draw reselects the complete maximum; S21 reports both prompts, AUROC and paired reductions.

TABLE 14. E12 ORACLE LOCALIZATION UNDER THE 11-ATTACK UNION. STANDARD PROMPT: 345 CROSS-FITTED EPISODES, 53 CUSTOMERS. MAXIMUM NETWORK BA (%) WITH CUSTOMER-BOOTSTRAP INTERVALS.

Target	Full trace	Writes removed	Writes only
Cancel	91.7 [82.1, 100.0]	62.8 [54.6, 75.3]	91.7 [82.1, 100.0]
Return	85.5 [76.7, 93.4]	59.9 [52.2, 71.8]	84.9 [78.8, 94.0]

7,639 requests. These results reject cover indistinguishability in this implementation.

Detecting the terminal flush is not identical to recovering the task category. A training-fitted detector cuts the predicted suffix before intent classification; Table 16 reports that composed attack. Its strongest return point is 60.7% under rules. The overall attack union retains both stripped and unstripped traces, so removing a suffix never replaces a stronger attack. Scheduling leakage is a concrete failure mode, not an unmeasured caveat.

Price the protected boundary. Serial/two-in-flight shadow schedules have real-chain p95 latency of 26.2/20.8 ms, versus 56.4/112.3 ms for the 2/5 ms tunnels. Every session preserves real responses and final live state, checking the designed logical invariant. This does not measure defended planner utility. Fixed tunnel cell sizes/counts still expose host-scheduled direction/order; no measured condition meets the two-target absolute criterion. Transport dummies protect the passive link without executing tools. A trusted gateway is not a reason to prefer expensive shadow execution on that link; dispatch protection addresses a different, restricted projection that excludes the executor's routing knowledge.

8.7. E14: matched attacks and classical controls

Seven added defense-aware models include a trained 1D-CNN, k-FP-style forest-leaf neighbors and tabular learners. Table 15 separates a common seven-model comparison from the strongest available union retaining every E13 attack. All fit on defended training customers; each bootstrap draw reselects the suite maximum.

Under the common suite, random-donor Walkie-Talkie (WT) gives 50.0/56.7% BA at 2.04× bytes, versus serial shadow's 50.0/52.9% at 15.11×: a trade-off, with no demonstrated network-efficiency advantage for shadow cover. WT also matches or improves both BA points, bytes and calls against parallel shadow. Under rules, WT gives 51.4/56.3% at 2.23×, improving those four point metrics over both shadow schedules (S23). These are descriptive point comparisons, not statistical dominance or equal-utility comparisons.

The WF rows are offline RPC adaptations, not reproductions of original website-fingerprinting guarantees. WT lacks symmetric site pairing; WTF-PAD uses training-IAT histograms rather than its Tor configuration. Input-derived fragments preserve sizes, directions and order; this is not a decryption or live-planner utility check. Offline schedule spans and measured RPC latency have different origins. S20 supplies all algorithms, both prompts and common-suite intervals. The evidence favors cheaper transport alternatives on some measured points and supplies no minimum-cost or absolute-security claim.

TABLE 15. TIMING-AWARE E13/E14 COMPARISON: STANDARD PROMPT, 138 HELD-OUT EPISODES, 24 CUSTOMERS (5 CANCEL-POSITIVE, 7 RETURN-POSITIVE). EACH BA PAIR IS CANCEL/RETURN (%). THE COMMON SEVEN MODELS PERMIT A MATCHED-SUITE COMPARISON; THE FULL UNION ALSO RETAINS ALL EARLIER ATTACKS. ALL FULL-UNION TWO-TARGET GATES REMAIN UNCONFIRMED.

View	Condition	Common-seven BA	Full-union BA	Upper BA	Bytes×	Calls×	RPC p95 (ms)
M	Undefended	84.3/76.9	91.7/86.4	100.0/97.2	1.00	1.00	15.4
M	Per-call constant	53.2/59.1	68.6/62.7	92.8/79.5	3.92	1.00	16.6
M	Serial shadow	50.0/52.9	61.4/53.4	71.4/65.8	15.11	3.83	26.2
M	Two-in-flight shadow	50.0/57.3	51.7/57.7	69.1/69.7	15.11	3.83	20.8
M	Full tunnel, 2 ms	59.9/55.2	62.5/55.2	70.2/62.0	125.71	1.00	56.4
M	Full tunnel, 5 ms	54.6/53.7	54.6/57.3	72.9/64.2	125.71	1.00	112.3
S	Record buckets	53.1/69.8	53.1/69.8	78.5/86.9	2.25	1.00	–
S	Tamaraw 40/12 ms	55.3/55.3	55.3/55.3	68.7/64.4	12.15	1.00	–
S	Tamaraw 4/1.2 ms	57.8/51.8	57.8/51.8	71.8/61.8	11.99	1.00	–
S	FRONT	59.0/69.3	59.0/69.3	69.9/81.8	2.51	1.00	–
S	WTF-PAD adaptation	51.9/62.7	51.9/62.7	64.1/79.6	1.88	1.00	–
S	WT random-donor	50.0/56.7	50.0/56.7	67.0/67.8	2.04	1.00	–
S	FRONT, short calibration	52.4/56.0	52.4/56.0	61.3/71.1	2.33	1.00	–

M: measured local tool RPC on saved actions; S: offline traffic transformation. RPC p95 ends at the last real response, excluding LLM/WAN. S-row simulated spans are not RPC latency. Upper BA is the full-union 95% bound. M unions contain 17–22 attacks; S unions contain the common seven. Ratios use prompt-specific undefended captures; S20/S23 give both prompts, interval and finish-time details. No row measures live defended-planner utility.

TABLE 16. E13 INTENT BA AFTER LEARNED FLUSH REMOVAL (95% CUSTOMER-CLUSTER INTERVAL). EACH ROW: 138 TEST EPISODES, 24 CUSTOMERS. MAXIMUM OF THREE CLASSIFIERS PLUS MAJORITY; THE COMPLETE ATTACK UNION ADDITIONALLY KEEPS UNSTRIPPED OBSERVATIONS.

Prompt	Schedule	Cancel	Return
Standard	Serial	57.8 [50.0, 70.7]	53.4 [50.0, 64.6]
Standard	Parallel	50.0 [50.0, 64.0]	50.0 [50.0, 56.7]
Rules	Serial	50.1 [50.0, 55.6]	60.7 [50.8, 70.9]
Rules	Parallel	50.8 [50.0, 61.7]	54.0 [50.0, 67.0]

8.8. E15: an additional model on the same environment

A separate 150-run GPT-6 Luna/low study uses 75 existing retail tasks, two prompts and one repetition (ten pilots excluded). Standard-prompt dispatch BA is 87.5/66.6% for writes versus 50.0/57.0% for get/find/list reads. Rewards are 32/75 and 33/75; seven protocol-error episodes remain included. Each prompt's attacker has only 29 training episodes. This repeats the operation-label diagnostic with a same-provider alias; it is not a second environment or semantic-attribute validation. Padding matches or improves joint network point estimates. S22 gives all cells and 600 TLS replays.

E16: a target independent of operation type. An exploratory attribute check fixes single-item returns and a pre-fit digital-device product grouping. Each prompt yields 20 customer–order–item instances from only 15 customers; repeats are aggregated and the original customer folds retained. The fixed LR/RF/GB suite gives 30.3% pooled OOF BA [16.7,52.7], versus 50% for a constant predictor. A single fixed shuffled target gives 65.2%; it is not a permutation test. Both prompt arms have identical features. Below-chance scores can reflect reverse associations; this small diagnostic neither confirms non-action-attribute inference nor proves no leakage (S23).

8.9. What the cases establish

The experiments test the comparison obligations. E5 ties protection to the emitter; E6/E7 tie cover to state and the final network view; E9–E12 locate operation-category signatures and price training-calibrated shaping. E13 exposes the clock and tool-response cost. E14 tests stronger learners and established WF mechanisms on the same saved workload; E15 checks a separate model alias. The supported outcome is a measured trade-off under explicit observation and execution contracts, not an absolute security bound or a novel padding construction.

9. Implications, Ethics and Limits

9.1. Deployment paths and reporting obligations

A passive agent–tool observer can be an access-network operator, an authorized VPC flow monitor, or a relay on an inter-organization path. These are access assumptions, not a measured prevalence estimate. Table 17 states the corresponding visibility and trusted endpoint. Encrypted ClientHello hides the inner server name when supported, but destination IPs and downstream connections can still identify tools [73]. The single-ingress experiments establish no multi-endpoint anonymity.

Shadow state has three possible implementations. **(1) Authorized reads and verified no-ops** use the backend with discarded responses; idempotence alone does not preserve state. **(2) Provider-supported isolation** includes Kubernetes server dry-run, separate Stripe sandboxes and transactional rollback [74], [75], [76]. These have different semantics: for example, sequence changes can survive rollback [77]. **(3) A trusted tool gateway** authenticates cover markers and routes shadow writes to an isolated executor. Only this local

TABLE 17. DEPLOYMENT BOUNDARY: THE RELEVANT PASSIVE OBSERVER AND WHAT THE MEASURED MECHANISM CAN PROTECT.

Setting	Visible information	Applicability / trusted point
Same-host sidecar	Host trace/process/IPC metadata, if permitted	Link padding alone does not hide host instrumentation; isolate the monitor's access.
Same-VPC gateway	Ingress record sizes, times and addresses	Tunnel or shadow shaping can protect this link; gateway/markers remain trusted.
Remote SaaS/MCP	IP, unprotected SNI, per-endpoint flows and times	Shared ingress needed for tool-identity cover; ECH alone does not hide destination IP.
Cross-organization	Each hop's routing; provider calls/commit log	No guarantee across all hops. A trusted isolated executor is an explicit added assumption.

gateway design is measured; the provider-specific paths are feasibility examples, not backend validations.

The trusted gateway/executor can identify cover. Protection ends at exposed markers, routing branches or commit logs; it does not protect against that principal. On the ingress link, the inexpensive WF adaptations are relevant competitors. At a restricted dispatch view, hiding call profiles additionally requires marker/routing isolation. Thus our results do not justify application-layer dummy execution for every passive-link deployment. Cover also consumes quotas and shared capacity; admission budgets, backpressure and separate credentials must prevent overload or accidental live writes.

9.2. Ethics and artifact scope

Executions use cloned public benchmark databases and loopback sockets; model requests contain public tasks and simulated interactions. No production account is modified or third-party service probed for a vulnerability. Stakeholders include paying tenants, shared-capacity providers and people affected by state changes. Other-account cancellation is a simulator control. Released inference code can support surveillance; examples use public/synthetic identities and controlled endpoints. Bot-detection evasion has a different authorization boundary [12], [6].

AI assisted candidate-field extraction, implementation, source reconciliation and prose revision. This supports a large, source-linked comparison; it supplies no independent human judgments. Authors' source verification is specified in §2.3. Saved-output replay, model-generated trajectories and measured local TLS are distinguished throughout. The artifact archives source locators, decisions, observations, calibration and executable checks. An anonymous submission snapshot must omit identity metadata and development history; the separate submission checklist records the AI-use disclosure and author review required at registration.

9.3. Threats to validity and our own evidence audit

The purposive corpus has no field-wide recall estimate. Archived extraction–verification agreement is non-independent; blind recoding sheets support future audit. Constructed rule cases and mechanism checks assess implementation consistency, not independent decision accuracy. Applicability-sensitive denominators and unresolved source cells are reported separately (§6; S23).

Applying the same utility requirement to our experiments gives a precise boundary: undefended agent reward is measured; defended logical response/state conformance is checked; *live planner utility under defense is not established*. E9/E10/E15 inherit reward from fixed trajectories, and E13 measures local RPC clocks under Windows scheduling, excluding LLM generation, WAN transit, deadlines and replanning. Neither state equality nor planner-message equivalence fills that gap.

E6/E7 calibrate on known reference programs. E9 freezes calibration before held-out collection, but development could access all 115 tasks. E10–E14 and the new sensitivity analyses are exploratory known-cohort analyses, not new-customer confirmation. Their intervals condition on fitted attackers and omit retraining uncertainty. E9's held-out set has only 5/7 positive customers; the full cohort has 14/18. Repeated runs increase execution coverage, not independent customer support.

Sol and Luna are mutable aliases from one provider in one retail environment, each also used in a separate user-simulator context. The Luna comparison has only 29 training episodes per prompt. Fixed open weights, cross-vendor transfer, a second environment, independent human recoding and live replanning under cover require new data. The SoK's source-bound comparisons and measured transfer failures stand separately from these untested generalizations.

10. Conclusion

This SoK turns published claims into source-bound comparison records. It identifies failures that paper-level grouping obscures: protection lost at a new plaintext-aware emitter, read cover reused across a state-changing boundary, and costs or utility borrowed from a different mechanism. Our own measurements expose the same mistakes: action-defined labels limit what write localization establishes, preserved replay state does not measure planner utility, and inexpensive transport adaptations match costly shadow shaping on selected measured point metrics. The practical output is an inspectable map of justified and unjustified reuse, with matched measurements where available and precise missing obligations where deployment claims remain unsupported.

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Appendix A. Reproducibility and Planner-Visible Equivalence

Artifact organization. The archived reproducibility package groups the source ledger and family audit, transport experiments, stateful tool-dispatch experiments and real-agent studies. A machine-readable module index resolves these groups and their extensions to source files, predictions and verification entry points. The README supplies environment versions, checksums and commands. The complete 690-run cohort and earlier releases retain their original membership and results for provenance.

The source module contains all 82 family records, the 39-result ledger, all 180 audit cells, search reconciliation and coding columns. The real-agent module contains the 414/276 training/test split, frozen calibration, model decisions, sandbox scores, 2,760 TLS sessions and saved attack predictions. Offline reconstruction reproduces the fitted-attack results and intervals without additional model calls. The real-agent module also includes customer-cross-fitted analyses, training-fold operating-point selection and an oracle write-call ablation (E10–E12, Supplement S15). Their analytic TLS record model is checked against the measured sessions. E13 adds 4,140 live tool sessions, observed record-arrival timestamps, pre-test frozen attack models and complete tunnel controls (S19). The paired program contrasts reported in the current paper are a post hoc analysis of those saved predictions, using the same 2,000 customer-cluster draws and recomputing the suite maximum for each condition in each draw.

The robustness extensions archive every stronger attack, prediction and defense transform. E14 rechecks measured clock leakage and compares explicitly specified WF adaptations (S20); the primary E10 and E12 updates retain all prior attacks and add CNN/forest-leaf learners (S21). Exact local refitting, model reloading and relocated-path verification make these outputs executable without new model calls. This is computational verification, separate from external independent replication or human recoding. Source follow-up corrections are an explicit overlay on the historical audit, preserving its original columns and agreement statistics.

E15 adds a separate Luna cohort (S22). Its public raw-event archive, dependency hashes and executable checks link model decisions to sandbox rewards and local TLS replays. Recalculation uses the archived benchmark/tool sources and the public raw records without new model requests. The original Sol cohort, fitted calibration and earlier outputs remain unchanged.

Equivalence invariant and assumptions. Let s_t be live tool state and H_t the planner-visible sequence of messages, actions and tool responses. A real action a_t produces $(s_{t+1}, o_t) = T(s_t, a_t)$. The defended dispatcher forwards a_t unchanged and executes cover transitions only on a separate state s_t^{sh}, suppressing their responses. With the same initial live state, deterministic simulator tools and the same realized real actions, cover cannot change s_{t+1} or o_t. Induction on real-action steps gives

$$s_t^D = s_t, \qquad \pi_{\mathrm{plan}}(D(H_t)) = H_t.$$

Framing preserves application payloads and end-of-episode flushing adds only hidden shadow transitions. This is equality of the logical planner transcript. It assumes no timeouts, shared-state side effects or timing-dependent tool behavior; equality of independent model samples is not required. Beyond E13's local tool-loop measurements, a fully interactive deployment would additionally need to establish that wall-clock delay and shared resource effects do not change task execution.

The frozen dispatcher copies each real-call record verbatim into the defended stream. Executable checks match the real-call projection and original scoring fields across all 690 trajectories and four program conditions. Shadow execution and wire-byte accounting are separately reconstructed by the original full-cohort verifier. Reward equality therefore expresses the isolation invariant; call and byte ratios quantify the transport intervention on that fixed workload.

Reading the main tables. The corpus map reports signal-generation and receiving layers separately. The per-family audit distinguishes bound, partial, unreported, unverified and inapplicable evidence. These statuses preserve source scope: an unverified binding identifies a comparison obligation rather than declaring an experiment absent. In the real-agent table, point BA, paired BA reduction and the absolute finite-suite criterion answer different statistical questions; all intervals condition on the saved fitted attackers.